

AI Business Insights & Alert Agent (n8n Blueprint)

Overview

This project is an AI-powered business monitoring system built using **n8n + AI agents**.

It automates the process of:

- Collecting business data
- Analyzing performance using AI
- Detecting anomalies
- Sending actionable insights via email

Objective

To replace manual tracking and dashboards with an **automated AI-driven decision system** that delivers insights directly to the user.

Workflow Architecture

Trigger → Data Source → Data Processing → AI Analysis → Decision Logic → Notification

Workflow Breakdown

1. Trigger Layer

Type: Webhook Trigger (POST)

- Used to manually or programmatically trigger the workflow
- Acts as an entry point from UI or API

Input (optional):

```
{}
```

2. 📊 Data Source Layer

Node: Google Sheets (Read Rows)

- Fetches business data from a predefined sheet
 - Expected columns:
 - Date
 - Sales
 - Leads
 - Revenue
-

3. ⚙️ Data Processing Layer

Node: Function / Code Node

- Cleans and structures the incoming data
- Converts rows into a format suitable for AI processing

Output Example:

```
{
  "raw": [...],
  "formatted_data": "structured summary"
}
```

4. 🧠 AI Analysis Layer

Node: AI Agent (OpenAI / LLM)

- Analyzes business performance
 - Generates:
 - Key insights
 - Trends
 - Root causes
 - Actionable suggestions
-

5. ⚡ Decision Logic Layer

● Revenue Check

Condition:

```
(today_revenue / yesterday_revenue) < 0.7
```

👉 Detects >30% revenue drop

Leads Check

Condition:

today_leads < 30

 Detects **low lead pipeline**

Priority Flow

IF Revenue Drop
→ Send Revenue Alert
ELSE
IF Low Leads
→ Send Leads Alert
ELSE
→ Send Normal Report

6. Notification Layer

Node: Gmail

Sends 3 types of emails:

Normal Report

- Daily insights
 - Trends and recommendations
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Revenue Alert

- Triggered when revenue drops significantly
 - Includes AI-generated explanation + actions
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Leads Alert

- Triggered when leads fall below threshold
 - Focuses on pipeline issues
-

Example Scenarios

Scenario	Output
Stable data	Daily report
Revenue ↓ >30%	Revenue alert
Leads < 30	Leads alert

Tech Stack

- **n8n** – Workflow automation
- **OpenAI / LLM** – AI analysis
- **Google Sheets** – Data storage
- **Gmail** – Notifications

Data Flow Summary

```
Webhook Trigger
↓
Google Sheets (Read Data)
↓
Function (Format Data)
↓
AI Agent (Generate Insights)
↓
IF Node (Decision Logic)
↓
Gmail (Send Alerts/Reports)
```

Key Features

-  AI-generated business insights
-  Automated anomaly detection
-  Priority-based alerting system
-  No dashboards required
-  Fully automated reporting

Design Decisions

Why Google Sheets?

- Simple and accessible data source
 - No backend required for MVP
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Why Webhook Trigger?

- Enables UI/API integration
 - Allows manual or external triggering
-

Why Priority-Based Logic?

- Avoids multiple alerts
 - Ensures only the most critical issue is reported
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Limitations (Current Version)

- Single data source (Google Sheets)
 - No user-level data separation
 - Manual data management required
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Future Improvements

- Multi-user support (userId-based filtering)
 - Database integration (Supabase/PostgreSQL)
 - UI dashboard for input & visualization
 - Weekly/monthly analytics
 - WhatsApp/Slack alerts
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Use Cases

- Small business monitoring
 - Startup analytics
 - Sales performance tracking
 - Automated reporting systems
-

Final Outcome

This system acts as:

- 🧠 AI Analyst
- ⚡ Decision Engine
- ✉️ Automated Notifier

Delivering insights without requiring dashboards or manual analysis.

★ Conclusion

This workflow demonstrates how **AI + automation** can transform raw business data into **actionable intelligence** in real-time.

Built using n8n + AI Agents